Module 1
Introduction to Vector Databases and Chroma DB

Welcome to the Course
- IBM Product Spotlight: watsonx.data — Reading • 2 min
- Video: Course Introduction — Video • 3 min
- Course Overview — Reading • 5 min
- RAG and Agentic AI Professional Certificate Overview — Video • 5 min
- Reading: Helpful Tips for Course Completion — Ungraded Plugin • 5 min

Introduction to Vector Databases and Similarity Search
- Vector Database Concepts — Video • 6 min
- Reading: Vector Databases Versus Traditional Databases — Ungraded Plugin • 5 min
- Vector Database Types — Video • 9 min
- Applications of Vector Databases — Video • 6 min
- Reading: Similarity Search — Ungraded Plugin • 30 min
- Lab: Similarity Search by Hand — Ungraded App Item • 30 min
- [Optional] Interactive Lesson Recap Podcast (AI-Powered) — Ungraded App Item • 10 min
- Practice Quiz: Introduction to Vector Databases and Similarity Search — Practice Assignment • Grade: 100%

Exploring Chroma DB
- Chroma DB Key Concepts and Architecture — Video • 6 min
- Reading: Chroma DB Filtering — Ungraded Plugin • 15 min
- Reading: Similarity Search and HNSW in Chroma DB — Ungraded Plugin • 30 min
- Lab: Similarity Search on Text Using Chroma DB and Python — Ungraded App Item • 30 min
- [Optional] Interactive Lesson Recap Podcast (AI-Powered) — Ungraded App Item • 10 min
- Practice Quiz: Exploring Chroma DB — Practice Assignment • Grade: 100%

Module Summary and Evaluation
- Reading: Summary and Highlights: Introduction to Vector Databases and Chroma DB — Reading • 3 min
- Reading: Cheat Sheet: Introduction to Vector Databases and Chroma DB — Ungraded Plugin • 15 min
- Graded Quiz: Introduction to Vector Databases and Chroma DB — Graded Assignment • 21 min
- [Optional] Discussion Prompt: Meet and Greet — Discussion Prompt • 10 min

Module 2
Vector Databases for Recommendation Systems and RAG

Reading: Summary and Highlights: Introduction to Vector Databases and Chroma DB

6 XP • Advanced RAG System Implementation

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Congratulations! You have completed this lesson. At this point in the course, you know:

- That vector databases simplify data storage, organization, and retrieval of complex data types, including images, likes, sounds, text files, pattern data, map data, genomic information, and others
- That you can use vector databases for analysis tasks that group items, classify items, and suggest relationships among items
- That vector databases, an integral part of machine learning, and for data in diverse domains, offer high performance and scalability
- That vector databases store data as mathematical objects defined by size and direction
- That vector databases use distributed computing, indexing, and parallel processing techniques to quickly manage big datasets and process queries
- That a vector is an array of numerical values relating to different features or data attributes
- That vector-based database types include in-memory, disk-based, distributed, graph-based, and time-based databases
- That vector-based databases, speed performance for recommendation systems, social network analysis, knowledge graphs, graph analysis, and other complex tasks
- That dedicated vector databases use data structures, reversed indexes, product quantization, and locality-sensitive hashing (LSH) that provides scalability, delivers speed, and enhanced customization
- That popular database vector database vendors include FAISS, Annoy, and Milvus
- That databases that support vector search are either regular database systems or data processing frameworks that let you store data as BLOBs, arrays, or user-defined types
- That notable vendors that support vector search include SingleStore with its support for IBM watsonx.ai, Elasticsearch, PostgreSQL, MySQL, RedisAI, Apache MongoDB, and Apache Cassandra
- That organizations can make data-driven apps that work well and can be expanded to many areas using their strengths
- That you can use vector databases for image and video analysis, including feature aspects, similarity searches, and real-time data processing
- That vector databases use embeddings and cross-domain suggestions to enhance the completeness of recommendation systems
- That GPS, fleet management, and real-time traffic routing suggestions are built on vector database geospatial processing capabilities
- That vector databases enable marketing and social sites to scale resources to manage user profiles, deliver trend analysis, and optimize their Cloud and other hardware use
- How to perform similarity search manually
- That Chroma DB is a powerful vector database designed for various retrieval tasks
- That Chroma DB supports multi-modal retrieval, metadata filtering, vector search, and full-text search
- That deploying Chroma DB can be done using client-server architecture or standalone mode
- That Chroma DB integrates with popular frameworks and supports multiple programming languages
- That Chroma DB uses an advanced algorithm for approximate nearest neighbor search, which efficiently finds the approximately closest chunks to a query within a collection
- That Chroma DB is suitable for a wide range of applications, including recommender systems, document search, image retrieval, and AI-based chatbots
- How to perform similarity search using Chroma DB

Mark as completed

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